

ML-T2-090: Model Complexity Log

Learn Depth Academy - Track 2 Project Documentation

Phase 1: Problem Formulation

Inputs & Outputs:

Tabular numerical features representing physical characteristics, outputting a binary classification prediction.

Objective:

To systematically determine when additional model complexity provides a meaningful, justifiable benefit over strong, simple baselines.

Success Metric:

A balanced evaluation of predictive accuracy against engineering costs, specifically measuring inference latency and memory complexity.

Constraints:

The solution must rely entirely on zero-cost, open-source software and legally usable public data.

Experiment Log: Baseline Runs

Experiment ID: E01

Hypothesis /

Change:

Establish baseline expectations for model complexity on a standard tabular classification task.

Method:

Compared a simple Logistic Regression (max_iter=10000) against a complex Random Forest (n_estimators=100) using the Breast Cancer dataset.

Metric: Accuracy and Training/Inference Time.

Result:

Logistic Regression achieved 97.66% accuracy (0.6802s) while Random Forest achieved 97.08% accuracy (0.2098s).

Observation:

The simple model outperformed the complex model in accuracy, proving that added complexity does not automatically guarantee better predictive performance. Retaining both models for further complexity constraint testing.